



7TH GRADE LIFE SCIENCE — CHAPTER 2 REVIEW

Cells, Tissues & Body Systems • S7L2



The Cell as a System

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A **cell** is the smallest unit of life — but it works like a tiny factory. Each part, called an **organelle** ("little organ"), does one job. When every organelle does its job, the whole cell stays alive: it gets nutrients, grows, reproduces, builds materials, and removes waste.

⚠ **Cells ≠ atoms.** An atom is the smallest unit of *matter*; a cell is the smallest unit of *life* and is made of billions of atoms.



Animal Cell Organelles

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Organelle	Job
Nucleus	Control center; holds DNA & directs the cell
Cytoplasm	Jelly that fills the cell & holds organelles in place
Cell Membrane	Outer border; controls what enters & leaves
Mitochondria	Powerhouse; releases energy from food
Lysosome	Cleanup crew; breaks down waste & worn-out parts
Vacuole	Storage; holds water, food, & waste



Plant Cell Extras

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Plant cells have everything animal cells have, **plus three extras**:

Plant-only part	Job
Cell Wall	Stiff outer layer for shape & support — why plants stand up
Chloroplasts	Capture sunlight to make food (photosynthesis); hold green chlorophyll
Large Central Vacuole	Big water sac; full = firm plant, empty = wilting

🔑 **Photosynthesis:** sunlight + water + carbon dioxide → food + oxygen.



How Organelles Cooperate

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No organelle works alone. **Getting energy** takes teamwork:

membrane lets sugar in > cytoplasm carries it > mitochondria release energy > lysosome cleans waste

One job — four organelles cooperating. That's a **system** in action.



Levels of Organization

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Living things are built from simple → complex:

Cell > Tissue > Organ > Organ System > Organism

Level	What it is	Example
Cell	Smallest unit of life	One muscle cell
Tissue	Similar cells, same job	Muscle tissue
Organ	Different tissues together	The heart
Organ System	Organs with a shared task	Cardiovascular system
Organism	All systems together	A human

System	Main Job	Key Organs
Cardiovascular	Pumps blood (oxygen & nutrients)	Heart, blood vessels
Respiratory	Brings in O ₂ , removes CO ₂	Lungs, trachea
Digestive	Breaks food into nutrients	Stomach, intestines
Excretory	Removes liquid waste	Kidneys, bladder
Muscular	Moves the body & its parts	Muscles
Nervous	Sends signals; controls & senses	Brain, nerves
Immune	Fights germs & disease	White blood cells, lymph nodes

 **Systems interact!** Running uses respiratory + cardiovascular + muscular + nervous + excretory all at once. If one stopped, you couldn't run.

Quick Check — Cells & Systems

A. Animal or plant only? Write A or P-only.

1. Cell wall ____
2. Mitochondria ____
3. Chloroplast ____
4. Nucleus ____

B. Match the system to its job:

1. Removes liquid waste → ____
2. Carries oxygen in blood → ____
3. Fights germs → ____

I Can... (Self-Check)

- | | |
|--|---|
| ☐ Name the organelles & their jobs | ☐ Order the 5 levels of organization |
| ☐ List the 3 plant-only cell parts | ☐ Match each body system to its job |
| ☐ Explain how organelles work as a system | ☐ Give an example of two systems interacting |